# Homework 2 Solutions

## Overview

### Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
using Plots
using LaTeXStrings
using CSV
using DataFrames
using Roots
```

## Problems (Total: 30 Points)

### Problem 1 (5 points)

The systems diagram should look like this:

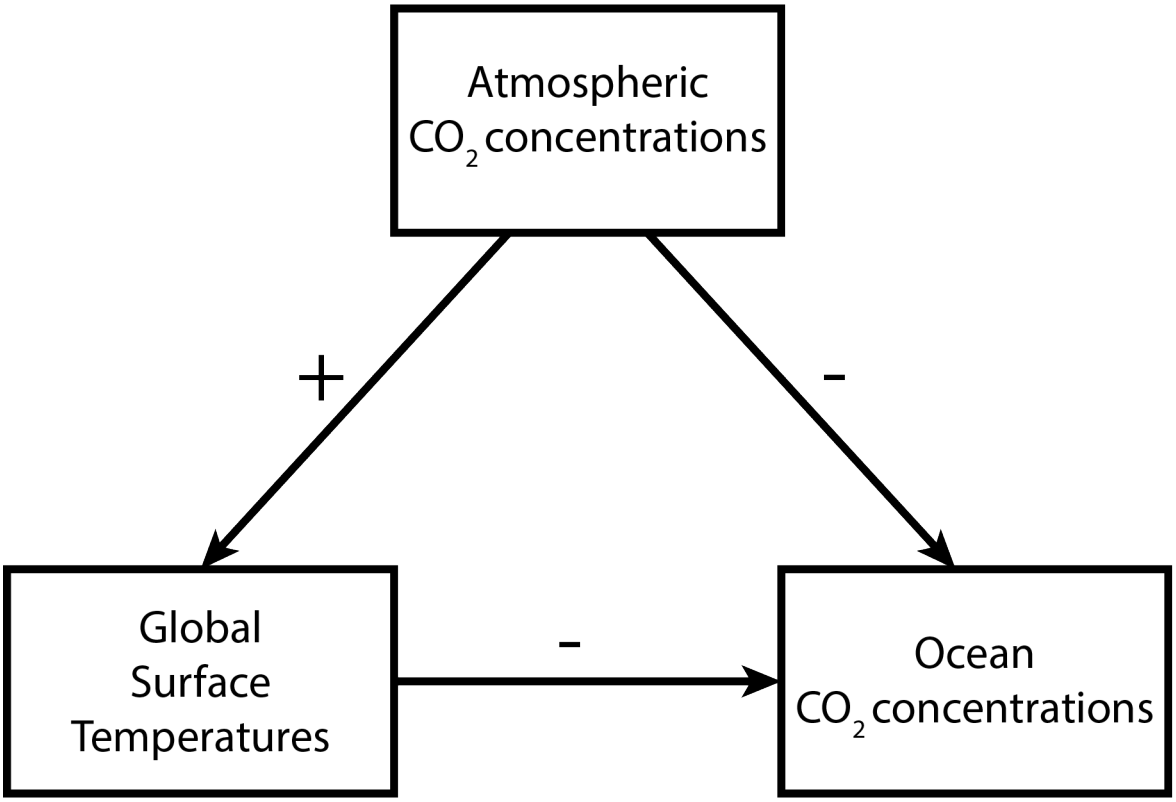


Figure 1: Systems Diagram for Problem 1

- The positive connection between atmospheric $CO_2$ concentrations and global temperatures is just capturing the greenhouse effect: $CO_2$ blocks the transmission of radiation along certain frequencies which would normally cause a loss of energy back to space.
- The negative connection between global temperatures and ocean $CO_2$ concentrations is a result of Henry's law for the solubility of $CO_2$. While the oceans are not an ideal mixture and the atmospheres and oceans aren't in equilibrium (so one would be cautious in using Henry's law to make specific predictions about concentrations), the key insight is the temperature dependence of Henry's constant, where increasing temperature decreases solubility.
- The negative connection between ocean and atmospheric $CO_2$ concentrations is the result of the mass balance of $CO_2$: $CO_2$ which degasses from the oceans must end up in the atmosphere, and the uptake of $CO_2$ from the oceans must come from the atmosphere.

The overall feedback is therefore amplifying: increases in temperature decrease the concentration of $CO_2$ in oceans, which increase the atmospheric concentration, causing further temperature increases.

**Problem 2 (15 points)**

**Problem 2.1**

This is a two box (or two control volume) model, as there is a change in the mass-balance at the second wastewater discharge. As a result, the relevant diagram might look like this:

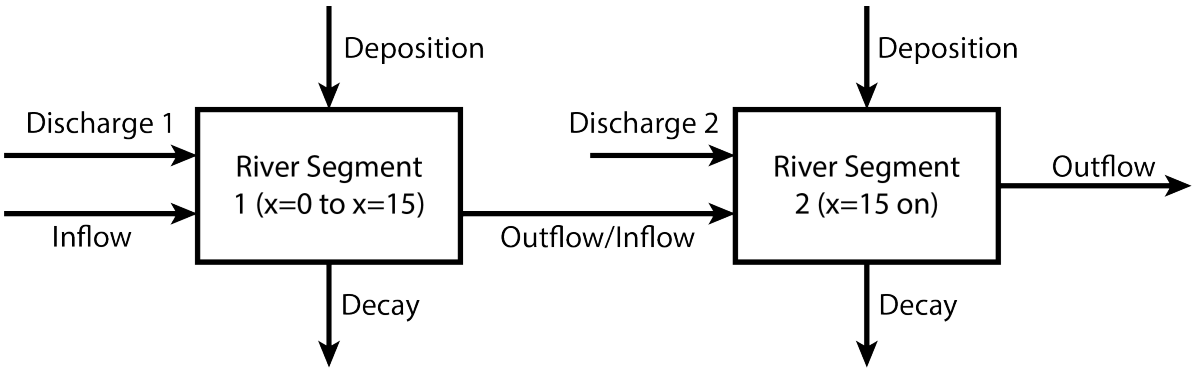


Figure 2: Schematic of the sources and sinks for the two-box river model.

**Problem 2.2**

CRUD decays exponentially at a rate of 0.36 $\mathrm{d}^{-1}$ and is deposited at a rate of 54 kg/km/d, so the differential equation for its mass $M$ as a function of time is

$$\frac{dM}{dt} = -0.36M + 54 \times 10.$$

By separating variables, the solution to this equation is

$$M(t) = M_0 \exp(-0.36t) + 1500\left(1 - \exp(-0.36t)\right),$$

where $M_0$ is the initial condition in each box. However, we need to convert time $t$ (d) to distance $x$ (km) as the discharges are constant in time and given in terms of spatial distance rather than time. As the velocity of the river is 10 km/d,

$$t = \frac{x}{10},$$

so

$$M(x) = M_0 \exp(-0.36x/10) + 1500\left(1 - \exp(-0.36x/10)\right). \tag{1}$$

We need to find the initial conditions at the entry to each box. The first segment goes from $x = 0$ (the first wastewater discharge) to $x = 15$ (the second discharge), and the initial condition satisfies $M_0 = M(0)$. To calculate this mass, we have to compute the total mass of CRUD in the combined inflow and wastewater discharge, which is

$$
\begin{aligned}
M_0 = M(0) &= (250,000\text{m}^3)\,(0.5\text{kg}/(1000\text{m}^3)) + (40,000\text{m}^3)\,(9\text{kg}/(1000\text{m}^3)) \\
&= 485\text{kg}.
\end{aligned}
$$

The second segment starts at $x = 15$ and follows the equation

$$
M(x) = M_1 \exp(-0.36(x-15)/10) + 1500\,(1 - \exp(-0.36(x-15)/10))\,, \qquad x \leq 15.
$$

The outflowing mass of the first box (evaluating Equation 1 at $x = 15$) is approximately $909kg$, so the initial condition at the second box is:

$$
\begin{aligned}
M_1 = M(15) &= 908\text{kg} + (60,000\text{m}^3)\,(7\text{kg}/(1000\text{m}^3)) \\
&= 909\text{kg} + 420\text{kg} \\
&\approx 1329\text{kg/d}.
\end{aligned}
$$

$$
M(t) = \begin{cases} 485\exp(-0.36x/10) + 1500\,(1 - \exp(-0.36x/10))\,, & \text{if } 0 \leq x < 15 \\ 1329\exp(-0.36(x-15)/10) + 1500\,(1 - \exp(-0.36(x-15)/10))\,, & \text{if } x > 15. \end{cases} \tag{2}
$$

### Problem 2.3

To see if the river will be in compliance with a standard of 2.3 kg/(1000m$^3$), we need to find the maximum value of Equation 2.

Plotting concentration over distance (we need to evaluate a bit downstream from the second discharge to be sure the atmospheric deposition won't cause the river to go out of compliance before the CRUD in the river can decay sufficiently, so we'll plot from $x = 0$ to 25 km):

```
# this function will return concentration in terms of kg/(1000 m^3)
function yuk_concentration(x)
    if x < 15
        M = 485 * exp(-0.36 * x / 10) + 1500 * (1 - exp(-0.36 * x /10 ))
        C = M / 290
    else
        M = 1329 * exp(-0.36 * (x - 15) / 10) + 1500 * (1 - exp(-0.36 * (x -
↪   15) /10 ))
        C = M / 350
    end
```

```
    return C
end

x = 0:25
yuk = yuk_concentration.(x)
p = plot(x, yuk, color=:black, linewidth=2, label="YUK")
hline!(p, [2.3], color=:red, linestyle=:dash, label="Standard")
xlabel!(p, "Distance (km)")
ylabel!(p, "Concentration (kg/1000 m³)")
```

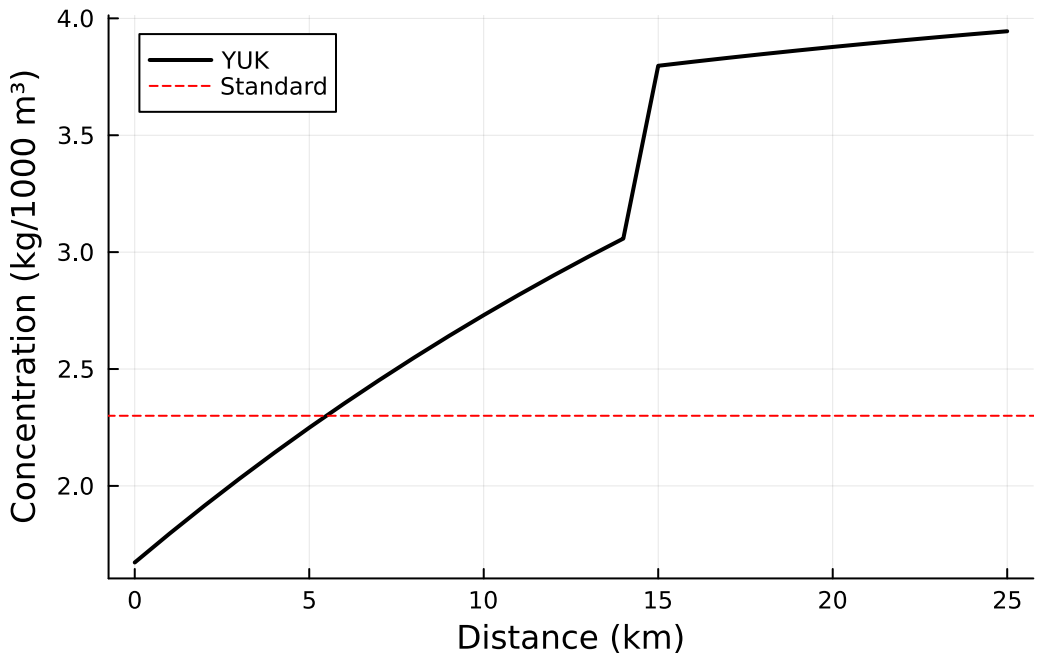


Figure 3: Concentration of YUK along the river.

We can see that the river is not in compliance.

### Problem 3 (10 points)

### Problem 3.1

The climate model (neglecting the greenhouse gas term) is:

$$C\frac{dT}{dt} = \frac{(1-\alpha)S}{4} - (A - BT) \tag{3}$$

Discretizing with forward Euler involves replacing $\frac{dT}{dt}$ with $\frac{T(t+\Delta t)-T(t)}{\Delta t}$ for some sufficiently small time step $\Delta t$. With this substitution and rearrangement, Equation 3 becomes:

$$T(t+\Delta t) = T(t) + \frac{\Delta t}{C}\left(\frac{(1-\alpha)S}{4} - (A - BT(t))\right) \tag{4}$$

We will select $\Delta t = 1$ yr, so Equation 4 becomes

$$T(t+1) = T(t) + \frac{1}{C}\left(\frac{(1-\alpha)S}{4} - (A - BT(t))\right) \tag{5}$$

### Problem 3.2

Implementing the model (including the ice-albedo feedback):

```
# function for the temperature-dependent albedo
function albedo(T)
    if T < -10
          = 0.5
    elseif T > 10
          = 0.3
    else
          = 0.5 + (0.3 - 0.5) * ((T + 10) / 20)
    end
    return
end

# incoming radiation as a function of temperature T
# (for albedo) and solar constant S
function incoming_radiation(T, S)
    return (1 - albedo(T)) * S / 4
end

# outgoing radiation, using the coefficients A and B
# from the problem statement
function outgoing_radiation(T)
    B = -1.3 # in W/m^2/°C
    A = 221.2 # in W/m^2
    return A - B * T
end

# climate model simulation as a function of number of years N,
```

```
# solar constant S, and initial temperature T
function climate_model(N, S, T )
    C = 51.0 # heat capacity, J/m^2/°C
    T = zeros(N+1) # pre-initialize storage for temperatures
    T[1] = T
    for t = 1:N
        T[t+1] = T[t] + (incoming_radiation(T[t], S) -
↪   outgoing_radiation(T[t])) / C
    end
    return T
end
```

Now we generate a number of simulations for different initial values and $S = 1272$ W/m$^2$:

```
init_T = -60:1:30
T = map(x -> climate_model(200, 1272, x), init_T)                    ①
p = plot(T, label=false, xlabel="Year", ylabel="Temperature (°C)")
```

① `map` is similar to broadcasting: it evaluates a function (in this case, `climate_model` over all values of an array (`init_T`)). We could also have done this with a loop.

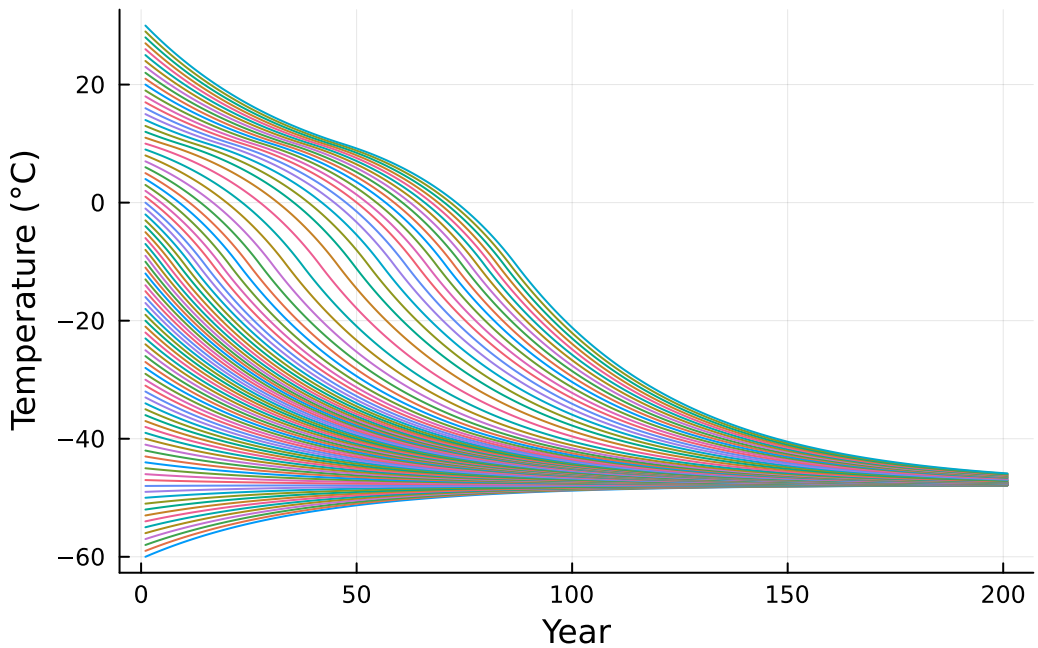


Figure 4: Temperature trajectories with the Neoproteorozoic solar constant for varying initial conditions.

We can see that with this solar constant, under this model the planet will inevitably tend towards a frozen equilibrium of approximately $-44°\text{C}$ (it's ok for this to not be exact so long as it's consistent with the plot). This is the only equilibrium and it is stable.

### Problem 3.3

If an increase in the solar constant was sufficient to warm the planet, we should see a simulation beginning with the Neoproteorozoic equilibrium of $-44°\text{C}$ warm to a non-frozen temperature with the modern solar constant. In reality, this transition would be slower than just a sudden change in the solar constant to the modern value, but if this simulation doesn't result in the necessary level of warming, a slower change in the solar constant would not be able to explain the transition. We can see if this works by using our climate model with the modern value of $S = 1368\ \text{W/m}^2$ and $T_0 = -44°\text{C}$.

```
T = climate_model(1000, 1368, -44)
p = plot(T, label=false, xlabel="Year", ylabel="Temperature (°C)")
```

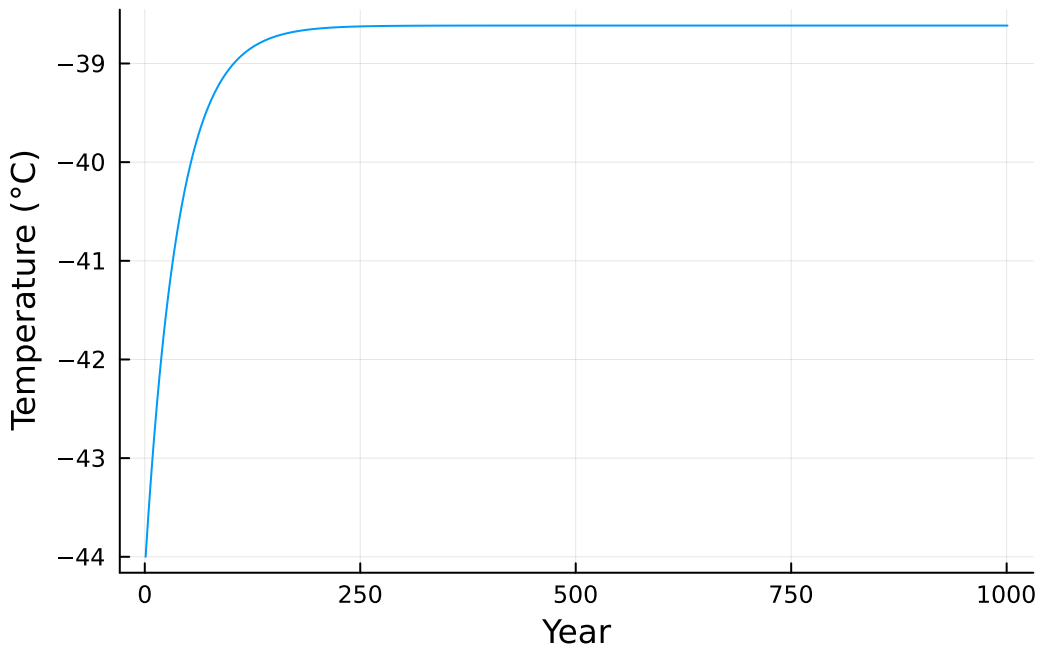


Figure 5: Temperature simulation from the frozen Neoprotereozoic equilibrium as the initial condition using the modern solar constant.

We can see that while this will result in the planet warming slightly (to around $-39°$C), it would not be enough to melt the ice covering the planet. This would require some other shock, like a change in $CO_2$ emissions (remember that we neglected the greenhouse effect), which could have come from *e.g.* volcanic eruptions.

## References

List any external references consulted, including classmates.